PPS Previous Year Questions

Unit-1

Introduction to programming

- ✓ Explain about computer components in detail?
- ✓ Write the difference between object code and executable code?
- ✓ Write and explain the structure of C program.
- ✓ What are the steps in program development?
- ✓ Differentiate between compiler and interpreter.
- ✓ Convert 11011101 to decimal system.
- ✓ Find the octal and hexadecimal equivalent of decimal number 235 and 25.
- ✓ Write a short note on creating and running the program. Bring out the importance of each step with using example.
- ✓ Explain about computer languages in detail.
- ✓ Explain about computing Environments in detail.
- ✓ Explain the steps for developing a software.

Algorithm & Flowchart

- ✓ What is an algorithm? What is its role in program development?
- ✓ Distinguish between algorithm and flowchart?
- ✓ Define an algorithm? Write an algorithm to find the sum of individual digits in a given number?
- ✓ Write an algorithm for determining the remainder of a division operation where the dividend and divisor are both integers?
- ✓ Write an algorithm to find the factorial of a given number.
- ✓ Construct a flowchart to identify largest number among the three given numbers.
- ✓ Draw a flowchart to find the roots of the quadratic equation.
- ✓ What are the various symbols used in Flowchart.
- ✓ Write an algorithm to print the numbers that do not appear in the Fibonacci series.
- ✓ Define flowchart and algorithm, distinguish between them.
- ✓ Draw a flowchart for printing sum of even terms contained within the numbers 0 to 20.

Data types and Variables

- ✓ What is a data type? Give the list of basic data types in “C” language and explain.
- ✓ What are the various data types? Give its size and the need for type casting.
- ✓ Explain different primary data types used in C language.
- ✓ Define variable.
- ✓ What is a variable? List the restriction on variable names in C. Give exampels?
- ✓ Differentiate between declaration and definition of a variable.

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- ✓ Discuss the scope of a variable with exampels?
- ✓ Distinguish between variables and constants. Give example.
- ✓ Explain in detail about C-Tokens.
- ✓ Distinguish between a keyword and a reserved word.
- ✓ What is an Identifier? List the rules required to form variable names in C.
- ✓ State and explain various identifiers in C program.
- ✓ Explain about type conversion mechanism available in C language.
- ✓ Write a C Program to swap two numbers.
- ✓ Explain a syntax for printf() function.
- ✓ Explain with examples, the various constants available in C language.
- ✓ What is type conversion? Explain two types of conversion with example?

Unit-2

Operators

- ✓ With examples briefly explain about Arithmetic operators.
- ✓ Explain about different types of operators in C.
- ✓ What is a conditional operator?
- ✓ Demonstrate the usage of relational operators.
- ✓ Demonstrate the usage of logical operators.
- ✓ Define bitwise operators and give examples.
- ✓ Explain about operator precedence and operator associativity
- ✓ Discuss about operator precedence in expression evaluation with a suitable example.

If_else, Switch

- ✓ What is a Selection Statement? Explain various selection statements in ‘C’ Language with suitable examples.
- ✓ Demonstrate nested if-else statements in “C” language using an example.
- ✓ Write a program showing the use of if else and switch statements in C?
- ✓ Briefly explain about switch statement.

Loops

- ✓ With the help of flowchart and example program explain the while loop.
- ✓ Distinguish among while, do-while and for statements in “C” language.
- ✓ Differentiate between While and Do While with an example.
- ✓ Explain the syntax of ‘for’ loop with an example.
- ✓ Using for loop write a program to print first n natural numbers.

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- ✓ Using while loop write a program to read 10 numbers from user and display their sum.
- ✓ Write a 'C' program to find whether the given number is prime or not.
- ✓ Explain the usage of break, continue and goto statements with a sample C program.
- ✓ Explain the difference between break, goto and continue statements.
- ✓ Write a C Program to demonstrate the break statement.
- ✓ Demonstrate the usage of break and continue statements using an example.
- ✓ Write a C program that reverses a number that is entered by the user by making use of do-while loop. How is this looping technique different from while loop?
- ✓ Write a program to find reverse of a number using while loop?
- ✓ Write a program to find whether the given number is palindrome or not.
- ✓ Write a program to find sum of n natural numbers using for loop?
- ✓ Write a program to print prime numbers up to a given number 'n' (Let n=100).
- ✓ Write a program that prints the binary equivalent of a decimal number.
- ✓ Write a c program to add prime numbers of a certain range (0 to 10).

Functions

- ✓ Define a function and explain the characteristics of a function.
- ✓ Distinguish between a function declaration and definition.
- ✓ Explain different elements of user defined functions in detail.
- ✓ Explain different ways of passing arguments to a function with an example.
- ✓ Differentiate between local and global variable.
- ✓ Give an example C program to show how an array can be passed to a function.
- ✓ Explain how arrays are passed to a function with an example.
- ✓ Explain about storage classes in C.
- ✓ What are the four storage classes? Discuss their scope rules.
- ✓ Explain about 'auto' storage class.
- ✓ What does extern mean in function declaration?
- ✓ Write a function to find the sum of digits of a given number.
- ✓ Define a function for determining whether a given character is a vowel or not.

Recursion

- ✓ Discuss about recursive functions. Write a C program to find factorial of a given number N by using recursive function.
- ✓ Write a program to display prime numbers in Fibonacci series using recursion.
- ✓ In what way a recursive function differs from non-recursive functions? Explain.

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Unit-3

Preprocessors

- ✓ What are preprocessors? Give an example.
- ✓ Describe the Pre Processor commands.
- ✓ What is pre-processor directive? Explain #define and #include pre-processor directives.

ARRAYS

- ✓ Define array?
- ✓ Define an Array. What are the types of array, write the compile time initialization syntax for 1D.
- ✓ What is an array? Explain run time initialization of an array?
- ✓ What is an array? Explain about initialization, accessing, and printing the array elements.
- ✓ Write a C program to find both the largest and smallest number in the given list of integers.
- ✓ Discuss array applications?
- ✓ Explain in detail about 2-D arrays with an example program
- ✓ What is an array? How to initialize two Dimensional array?
- ✓ Explain the concept of Arrays to function using program.
- ✓ Write a C program for addition of two matrices.
- ✓ Write a c program on addition of two matrix using Functions.
- ✓ Write a c program on matrix multiplication using Function
- ✓ Write a C program to find multiplication of Two Matrices.
- ✓ Write a C program to find product of n*n matrices?
- ✓ What is an array? What is subscript in an array? Give example to declare single array, 2D and multi-dimensional array.
- ✓ What is the difference b/w two-dimensional and multi-dimensional array?
- ✓ Explain bubble sort with an example.
- ✓ Write an algorithm for linear search.
- ✓ Explain linear search algorithm with suitable example.
- ✓ Write a program to find the second maximum in the array using functions.
- ✓ Write a c program to read N numbers into an array and perform binary search.
- ✓ Write an algorithm and develop a c program that reads N integer numbers and arrange them in ascending order using selection sort.
- ✓ Write a program that uses a function to sort the elements given by the user using selection sort.
- ✓ Write the algorithm for binary search.
- ✓ Write a program that uses a function to search a number within an array.
- ✓ Write a program to find the second maximum in an array using function.

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Unit-4

Strings

- ✓ What is String?
- ✓ Discuss the library functions for string to data conversion.
- ✓ What are the string manipulation functions? Explain any three of them.
- ✓ Write a program to find the length of the given string.
- ✓ Write a program to concatenate two strings.
- ✓ Write a "C" program to find length of a string and concatenation of two strings without using string handling functions.
- ✓ Write a 'C' program to append a string to the end of other string by using string manipulation function.
- ✓ Write a program to check whether a given string is a palindrome or not.
- ✓ Write a C program that reads a sentence and prints frequency of each of the letter in a given sentence.
- ✓ Briefly explain about getchar() and putchar() functions.

Pointers

- ✓ What is pointer? Write its applications?
- ✓ What is pointer to pointer?
- ✓ Write a c program to illustrate the use of indirection operator to access the value pointed by pointer.
- ✓ What is a pointer? Write a c program to find the sum and average of all elements in an array using pointer.
- ✓ What is Lvalue and Rvalue? Explain about pointers to void?
- ✓ What is pointer variable? Give example? And what are the arithmetic operations that can be performed?
- ✓ Write short notes on pointer arithmetic operations? What are the specific advantages of these?
- ✓ Why pointers should have data types when their size is always 4 bytes (in a 32 bit machine), irrespective of the variable they are pointing to?
- ✓ Write short notes on call by reference.
- ✓ Difference between call by value and call by reference with an example.
- ✓ Bring out with examples, differences between array name and pointer.
- ✓ Is null pointer same as uninitialized pointer.
- ✓ What is meant by dynamic memory management? How C language supports programmers in memory management? Explain with relevant code segments.
- ✓ What is command line argument?

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Unit-5

Structures

- ✓ Explain about structures?
- ✓ Define a structure for Bank Account. Implement it with basic data members and member methods.
- ✓ Declare a structure to store the following information of an employee- employee code employee name, salary, department number. Write a c program to store the data of n employees where n is given by user and use a function to display the employee information getting the maximum and the minimum salary.
- ✓ How do you define structure within a structure? Explain with an example.
- ✓ What is self-referential structure? Give an example.
- ✓ Give the description of self-referential structures.
- ✓ What do you mean by union? Give example.
- ✓ Differentiate structure and union with an example.
- ✓ Which one take less memory, structure or union. And why?
- ✓ How a structure data type is different from an array? Explain with an example.
- ✓ Explain the difference between array and structures. When should each of these be used?
- ✓ What is bit field in structures?
- ✓ Write short notes on Linked List.
- ✓ Explain about typedef?

Files

- ✓ What is a file? What are file operations?
- ✓ Explain the difference between binary file and text file? Where is each of them used?
- ✓ How files are accessed randomly?
- ✓ In how many modes a file can be opened? What are they?
- ✓ Write the syntax for opening a file with various modes and closing file.
- ✓ Write syntax for fopen() and fclose().
- ✓ Write a program to read from a file.
- ✓ Write about file handling operations in C.
- ✓ Write a program to copy contents from one existing file into another file.
- ✓ What are primary advantages of using a data file?